

Exit Card!

Complete the chart below.

Function	Type	degree	leading coefficient	End behavior
$y = x^2 - 2x$	Quadratic	2	1	As $x \rightarrow -\infty$, $y \rightarrow \underline{\infty}$ As $x \rightarrow \infty$, $y \rightarrow \underline{\infty}$
$y = -(x-1)(1-x)^2$	Cubic	3	-1	As $x \rightarrow -\infty$, $y \rightarrow \underline{\infty}$ As $x \rightarrow \infty$, $y \rightarrow \underline{-\infty}$
$y = -2x^5 + 7x^4 - 3x^3 - 18x^2 + 5$	Quintic	5	-2	As $x \rightarrow -\infty$, $y \rightarrow \underline{\infty}$ As $x \rightarrow \infty$, $y \rightarrow \underline{-\infty}$
$y = -(1-2x)^3(x+1)^2$ $= -(-2x)^3(x)^2 + \dots$ $= -(-8x^3)(x^2) + \dots$ $= 8x^5 + \dots$	Quintic	5	8	As $x \rightarrow -\infty$, $y \rightarrow \underline{-\infty}$ As $x \rightarrow \infty$, $y \rightarrow \underline{\infty}$
$y = -3x^2(2-x)^3(2x-1)^2$ $= -3x^2(-x)^3(2x)^2 + \dots$ $= 12x^7 + \dots$	polynomial	7	12	As $x \rightarrow -\infty$, $y \rightarrow \underline{-\infty}$ As $x \rightarrow \infty$, $y \rightarrow \underline{\infty}$

How am I doing?

